

Guide 33 - Machinist and CNC Machine Operator

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Purpose: Practical career-entry and advancement guide for machinists, CNC machine operators, setup technicians, and closely related precision-machining roles.

This guide is educational. It does not guarantee employment, wages, admission, funding, reimbursement, apprenticeship placement, certification, licensing, promotion, or legal authority to perform machining work. Requirements vary by jurisdiction, employer, machine, material, process, drawing, contract, and worksite. Verify current rules and qualified supervision before paying for training or performing hazardous work.

1. What this work is and who it may fit

Machinists and CNC machine operators turn drawings, specifications, models, and production instructions into precise physical parts. They work with lathes, mills, grinders, drills, boring equipment, electrical-discharge machines, measuring instruments, and computer numerical control systems. Some roles emphasize loading and running established CNC programs. Others include setup, tool selection, offsets, inspection, troubleshooting, process improvement, manual machining, or CNC/CAM programming.

This field can fit people who enjoy practical problem-solving, measurement, mechanical systems, computers, precision, and visible results. It rewards patience and disciplined checking. A small error in a dimension, tool offset, workholding setup, material identity, or program can scrap a part, damage tooling, injure someone, or create a serious quality problem.

A title alone does not define scope. “CNC operator,” “CNC machinist,” “setup machinist,” “manual machinist,” “tool-room machinist,” “CNC programmer,” and “manufacturing technician” may represent very different levels of responsibility. Read the actual job description and ask what work you are expected and authorized to perform.

2. What workers actually do

Typical duties may include:

- reading blueprints, sketches, engineering drawings, CAD/CAM files, work orders, process sheets, and specifications;
- identifying material and confirming part, revision, quantity, and routing information;
- selecting and installing approved tooling, fixtures, workholding, and measuring equipment;
- setting machine coordinates, work offsets, tool offsets, feeds, speeds, and related parameters within established procedures and training;

- loading and running proven CNC programs or, in more advanced roles, editing or creating programs under controlled processes;
- operating manual lathes, mills, grinders, drills, saws, or other machine tools where the job requires it;
- measuring parts with calipers, micrometers, indicators, height gauges, bore gauges, comparators, coordinate-measuring systems, or other approved inspection tools;
- monitoring tool wear, finish, chips, coolant, machine condition, cycle behavior, and dimensional drift;
- recording inspection results, deviations, nonconformities, tool changes, setup information, and production data;
- coordinating with programmers, engineers, quality inspectors, maintenance staff, production supervisors, welders, fabricators, and other manufacturing teams;
- stopping and escalating when the setup, drawing, machine condition, program, tolerance, guarding, inspection method, or safety condition is unclear.

Title distinctions matter

CNC machine operator often runs established programs and performs routine loading, monitoring, and checks.

CNC setup operator or setup technician typically takes greater responsibility for workholding, tooling, offsets, first-piece setup, and process readiness.

Machinist may combine manual and CNC setup, operation, troubleshooting, measurement, and process knowledge.

CNC programmer / CAM programmer creates or modifies machine instructions and process strategies and generally requires deeper manufacturing knowledge.

Do not claim setup, programming, inspection, tool-and-die, maintenance, or journey person competence unless you can support it with actual training and experience.

3. Safety and task boundaries

Machining is precision work around moving machinery, sharp tools, rotating parts, chips, coolant, heavy stock, electricity, stored energy, and sometimes automated material handling. Good machining practice begins with hazard control rather than speed.

Machine guarding and moving parts

OSHA states that machine parts, functions, or processes that may cause injury must be safeguarded. Point-of-operation hazards, rotating components, flying chips, belts, gears, couplings, and other moving parts can cause crushing, lacerations, entanglement, amputations, and eye injuries.

Never bypass or defeat guards, interlocks, door switches, enclosures, light curtains, emergency systems, lockout/tagout procedures, or approved machine controls to save time or recover production.

Rotating equipment and entanglement

Loose clothing, jewelry, long hair, gloves, rags, or hand tools can become entangled in rotating workpieces, chucks, spindles, cutters, or bars. PPE must match the task and employer procedure. A glove that protects against one hazard can create an entanglement hazard around rotating machinery.

Chips, tools, and workholding

Hot or sharp chips can cut or burn. Incorrectly clamped stock, fixtures, tools, inserts, or workpieces can move or be ejected. Use approved chip-removal methods and never reach into a hazard zone based only on the assumption that a spindle or axis has stopped.

Metalworking fluids

Metalworking fluids can expose workers through skin contact and mist or aerosol. OSHA notes possible skin, eye, nose, throat, and respiratory effects when fluids are poorly controlled or maintained. Follow the employer's SDS, fluid-management, ventilation, hygiene, splash-control, and PPE procedures.

Setup, maintenance, and troubleshooting

Changing tools, touching off, measuring in-machine, clearing jams, cleaning, recovering from alarms, adjusting workholding, changing fixtures, or troubleshooting can create additional hazards. Know when lockout/tagout, maintenance authorization, supervisor involvement, or a qualified technician is required.

Stop and escalate when

- the drawing, revision, datum, tolerance, material, process sheet, program, tool list, or setup instruction is unclear;
- the machine guard, enclosure, interlock, chuck, fixture, toolholder, coolant system, lifting device, or emergency control appears damaged or unreliable;
- the task would require bypassing a safeguard or reaching into a danger zone;
- a crash, unusual vibration, sound, smell, heat condition, tool failure, part movement, alarm, or unexpected machine motion occurs;
- a supervisor asks you to conceal scrap, skip inspection, falsify measurements, override a safety control, or run a process outside your verified competence;
- the work involves hazardous materials, unusual pressure/energy systems, export-controlled or defense data, medical-device traceability, aerospace requirements, or other regulated quality controls you have not been trained to handle.

4. Entry routes and transferable skills

The U.S. Bureau of Labor Statistics says machinists typically need a high school diploma to enter the occupation and usually receive long-term on-the-job training. Some learn through vocational schools, community or technical colleges, employer training, or apprenticeship programs.

Common routes include:

- entry-level machine operator or production roles with structured progression;
- paid employer training;
- advanced-manufacturing apprenticeship where an approved occupation and local program are available;
- public technical school or community-college machining/CNC certificates;
- military machining or manufacturing experience translated honestly into civilian requirements;
- SENA machining/CNC-related training in Colombia;
- maintenance, fabrication, welding, toolroom, inspection, CAD/CAM, industrial-mechanics, or manufacturing experience that transfers into machining.

Useful transferable skills include:

- arithmetic, fractions, decimals, geometry, trigonometry, and unit conversion;
- mechanical reasoning and spatial visualization;
- reading drawings and dimensions;
- careful use of measuring instruments;
- computer comfort and willingness to learn controller interfaces;
- documentation and quality-control discipline;
- safe tool handling and housekeeping;
- troubleshooting without improvising outside authorized procedures;
- communicating clearly when a dimension, setup, program, or process is uncertain.

5. United States labor-market evidence

Official national statistics

The U.S. Bureau of Labor Statistics reports for **machinists**:

- **May 2024 median pay: \$56,150 per year;**
- **lowest 10 percent: below \$38,100;**
- **highest 10 percent: above \$78,760;**
- **projected employment change, 2024–2034: approximately 0% for machinists specifically.**

For the combined machinists and tool-and-die-makers profile, BLS projects about **34,200 openings per year**, on average, over the decade, largely from replacement needs.

These are national occupational statistics. They are not apprentice, CNC-operator-only, programmer, tool-and-die, union, local-market, shift, overtime, or guaranteed wage figures.

Current non-government market estimate

Indeed's U.S. CNC Machinist salary page, updated **July 20, 2026**, reported an average base salary of approximately 28.06 *per hour**, *with a displayed range of about* ***20.44–\$38.53 per hour** and a methodology based on job-posting salary data.

This is a **non-government market estimate**, not an official federal statistic and not definition-equivalent to BLS data. Use it only as a current market cross-check.

When comparing offers, ask about shift differentials, overtime, production incentives, tool allowances, uniforms/PPE, training time, certification costs, health benefits, union terms, weekend schedules, travel, and whether setup or programming responsibilities are compensated differently.

6. United States training, funding, and free-first pathways

Before taking private-school debt, compare paid and lower-cost routes.

Apprenticeship and employer training

Apprenticeship.gov's Advanced Manufacturing resources identify CNC Machine Operator and Precision Machinist among apprenticeship occupations. At the same time, the current Apprenticeship.gov occupation page for O*NET Machinists 51-4041.00 states that that specific occupation entry is not currently approved for Registered Apprenticeship use.

That distinction matters. Do not assume every "machinist apprenticeship" advertisement has the same federal registration status. Verify the sponsor, registered occupation, wage progression, related instruction, work hours, tools, credentials, cancellation terms, and completion status for the exact program.

WIOA and American Job Centers

Workforce Innovation and Opportunity Act services can support occupational skills training, on-the-job training, incumbent-worker training, workplace instruction, skills upgrading, retraining, and customized training when local eligibility rules are met.

Ask your local American Job Center or workforce board:

- whether the machining/CNC program is on the applicable Eligible Training Provider list when required;
- whether you personally qualify before enrolling;
- whether an Individual Training Account or another training contract is available;
- whether tuition, books, tools, PPE, transportation, childcare, testing, or supportive services may be covered;
- whether approval must occur before you incur costs;
- what documentation and attendance/performance requirements apply.

Never assume WIOA will pay simply because a course is job-related.

Scholarships and public colleges

CareerOneStop's Scholarship Finder can help locate manufacturing-related scholarships, grants, and other awards. Sponsors can change deadlines and eligibility, so verify the current sponsor page before applying.

Public technical schools and community colleges may offer machining, CNC, industrial technology, metrology, CAD/CAM, or manufacturing programs at lower cost than private schools. Compare lab access, machine types, instructor time, safety controls, employer connections, placement claims, total fees, and whether the curriculum matches local hiring needs.

Employer support

Ask employers about paid trainee roles, incumbent-worker training, tuition reimbursement, credential reimbursement, tool allowances, uniforms/PPE, paid study time, shift flexibility, and advancement from operator to setup, machinist, inspection, or programming roles. Obtain written reimbursement terms before enrolling.

7. NIMS and credential boundaries

The National Institute for Metalworking Skills (NIMS) offers machining credentials in areas including Measurement, Materials & Safety; CNC Mill Operations; CNC Mill Programming, Setup & Operations; CNC Lathe Operations; CNC Lathe Programming, Setup & Operations; manual milling and turning; grinding; EDM; and advanced CNC/CAM roles.

NIMS states that most credentials combine a knowledge exam with a performance evaluation. Its current policy also states that credentials earned on or after **June 30, 2026** are valid for five years unless renewed under the current renewal policy; older credentials are not retroactively placed under that new expiration rule.

Important boundaries:

- NIMS is an **industry credential**, not a government license;
- one credential does not prove competence in every machine, controller, material, tolerance, inspection method, or programming environment;
- employers may use NIMS as a hiring or progression benchmark, but not every employer requires it;
- a school completion certificate is not automatically a NIMS credential;
- verify current fees, assessment requirements, renewal rules, and employer recognition before paying.

8. Canada pathway

Red Seal - Machinist

Canada's Red Seal program identifies **Machinist** as a Red Seal trade associated with **NOC 72100**. The occupational standard covers conventional, portable, and CNC machine tools including lathes, mills, grinders, drills, boring machines, EDM, and related equipment.

The national harmonization recommendation identifies **4 levels of technical training** and **7,200 total apprenticeship training hours**. Provincial and territorial authorities administer apprenticeship registration, certification, exams, compulsory/voluntary trade status, and local implementation. Verify the authority where you will train and work.

Government of Canada Job Bank wages

The Job Bank national wage page for CNC/general machinists currently shows approximately:

- **C\$21.00/hour low;**
- **C\$30.00/hour median;**
- **C\$41.50/hour high.**

The page states the wages were updated November 19, 2025 and uses a 2023–2024 reference period. These are national estimates, not guaranteed offers.

Apprentice financial support

Eligible apprentices in designated Red Seal trades may access the **Canada Apprentice Loan**, currently offering up to **C\$4,000 in interest-free loans per period of technical training**, subject to program rules. Quebec has alternate support arrangements.

Federal pages also identify Employment Insurance support during eligible technical training and other apprenticeship supports. Do not rely on outdated claims about the former Apprenticeship Incentive Grant or Apprenticeship Completion Grant as currently open 2026 funding; those grant programs closed for new progression/completion dates after March 31, 2025.

9. Colombia pathway

SENA machining and CNC training

Current SENA institutional records include **Mecanizado de productos metalmecánicos** and **Diseño de elementos mecánicos para su fabricación con máquinas herramientas CNC** within its industrial-mechanics training environment.

Use SENA's current training platforms to verify which programs are actually open, in which city or regional center, at what level, and with what entry requirements. Historical program names prove that a pathway exists but do not guarantee that a cohort is open today.

SENA Agencia Pública de Empleo

SENA's Agencia Pública de Empleo (APE) is a free public employment service. Current machining-related vacancies show employers requesting combinations of technical or technologist education, drawing interpretation, CNC setup/operation, metrology, and machining experience.

Use APE vacancies to compare real employer requirements and salary bands when supplied. A vacancy can expire quickly and is not a national wage standard.

Colombia work-boundary rule

Employer, sector, customer, quality-system, safety, export-control, and technical requirements can differ. Verify the actual workplace rules before assuming a foreign credential, private course, or prior experience automatically authorizes a particular machining task.

10. Latin America beyond Colombia

There is no single Latin American licensing, apprenticeship, wage, or training system for machinists and CNC operators.

For any country:

1. identify the public technical-training, labor, occupational-safety, and employment authorities;
2. check public vocational institutes before paying a private provider;
3. compare apprenticeship, dual-education, trainee, union, and employer-sponsored routes;
4. confirm whether credentials from another country or private provider are recognized;
5. verify local wage data separately rather than converting U.S. or Canadian figures into a supposed regional standard;
6. check rules for machine safety, lockout/tagout, industrial chemicals, working time, PPE, and quality systems in the actual jurisdiction;
7. verify whether the job involves regulated aerospace, medical, automotive, defense, energy, or other customer-specific requirements.

Do not transplant U.S. OSHA rules, Canadian Red Seal requirements, Colombia SENA pathways, or NIMS assumptions into another jurisdiction without verification.

11. Skills to build deliberately

Machining foundations

Develop competence in:

- fractions, decimals, tolerances, geometry, trigonometry, and metric/imperial conversion;
- reading orthographic drawings, section views, dimensions, notes, surface-finish requirements, and revision controls;
- material identification and traceability;
- workholding, fixturing, toolholding, cutter geometry, tool wear, and chip control;
- feeds and speeds as supervised manufacturing parameters rather than memorized universal numbers;
- manual lathe and mill concepts even if your first role is CNC-focused;
- offsets, datums, machine coordinates, work coordinates, and basic CNC control logic;
- dimensional inspection using approved instruments and calibrated equipment;
- coolant and metalworking-fluid awareness;
- safe part handling, deburring, cleaning, and shop organization.

CNC and digital skills

Build familiarity with:

- controller navigation and alarm interpretation;

- program structure and common G-code/M-code concepts without pretending every controller behaves identically;
- setup documentation and tooling lists;
- CAD/CAM workflow basics;
- simulation and verification;
- first-article and in-process inspection;
- file/version control and controlled transfer of programs;
- basic cybersecurity awareness for networked manufacturing equipment.

Professional skills

Develop habits in:

- punctuality and production reliability;
- documenting measurements honestly;
- communicating deviations early;
- protecting customer and employer drawings and data;
- asking before changing a controlled program or process;
- distinguishing scrap, rework, deviation, and acceptable product according to the quality system;
- learning from inspection results, setup errors, tool failures, and supervised problem solving.

12. A safe beginner learning sequence

A practical sequence is:

1. **Safety first:** guarding, emergency stops, rotating-equipment hazards, lockout/tagout awareness, chip handling, coolant, PPE, lifting, housekeeping, and reporting procedures.
2. **Measurement:** rulers, calipers, micrometers, indicators, tolerances, repeatability, units, and careful recording.
3. **Drawings and math:** views, datums, dimensions, fractions, decimals, geometry, trigonometry, and unit conversion.
4. **Manual machining concepts:** understand turning, milling, drilling, workholding, tooling, speed/feed relationships, and machine motion.
5. **CNC operation:** load a proven setup and program under supervision; understand controls, offsets, cycle start, feed hold, alarms, and inspection points.
6. **Setup:** progress to tooling, fixtures, work offsets, tool offsets, first-piece setup, and controlled verification.
7. **Programming/CAM:** only after understanding machining consequences, learn program creation/editing, simulation, post-processing, and controlled release.
8. **Advanced work:** multi-axis machining, tight-tolerance work, difficult materials, probing, EDM, automation, process planning, or specialized quality systems after the fundamentals are dependable.

Do not confuse quickly learning controller buttons with being ready to set up or program an unfamiliar machine safely.

13. A 90-day entry plan

Days 1–30: verify the occupation and local pathway

- Read several real local machinist/CNC postings and record repeated requirements.
- Visit or contact public training providers and ask what machines, controllers, lab hours, measuring tools, and employer partnerships are included.
- Compare paid trainee/apprenticeship routes with school-first routes.
- Ask American Job Centers, Canadian apprenticeship authorities, SENA, or the equivalent local public system about current support.
- Build or refresh shop math and measurement basics before buying expensive training.

Days 31–60: build demonstrable fundamentals

- Practice drawing interpretation and dimensional measurement.
- Learn safe machine concepts in a supervised lab or workplace.
- Document completed training and actual hands-on tasks without exaggerating scope.
- If using simulators, label simulator experience honestly and do not present it as production machining.
- Start a simple skills portfolio with non-proprietary practice drawings, inspection sheets, setup notes, or screenshots where permitted.

Days 61–90: target the first credible role

- Apply to roles matching your actual level: trainee, operator, apprentice, machine helper, production technician, junior CNC operator, or entry machinist.
- Tailor your résumé to the exact equipment, measurement, drawings, safety training, and coursework you can support.
- Prepare to explain how you handle uncertainty, inspection, safety stops, and mistakes.
- Ask employers how progression works from operator to setup, machinist, quality, or programming responsibilities.

14. How to evaluate a training program before paying

Ask for written answers to:

- What exact machines and controllers will I use?
- How many supervised lab hours are included?
- What percentage of the course is hands-on versus video or lecture?
- Which measuring instruments will I use?
- Does training cover both operation and setup, or only button-level operation?
- Is CAD/CAM included, and which software?
- Does the program prepare for an external credential such as NIMS, or only issue its own certificate?
- What are the total tuition, fees, tools, PPE, books, software, testing, and retake costs?

- Are placement rates independently documented, and how is “placement” defined?
- Which employers hire recent graduates from this exact program?
- Can I inspect the lab and speak with recent students?
- Is public funding or employer reimbursement available, and must approval occur before enrollment?
- What is the refund/cancellation policy?

Be cautious when a provider guarantees a job, promises an unusually high salary, implies that one short course makes you a programmer or journey person, hides lab access, pressures you to borrow immediately, or uses employer logos without clear evidence of an active relationship.

15. Building a truthful machining portfolio

A useful entry-level portfolio can include:

- shop-math or blueprint-reading exercises;
- non-proprietary practice prints;
- measurement records showing target, actual, tolerance, and instrument used;
- photos of practice parts when the school/employer allows it and no proprietary information appears;
- a simple process plan for a training part;
- screenshots from a CNC simulator or CAM training project clearly labeled as simulation/training;
- NIMS or other credentials actually earned;
- safety training certificates actually completed;
- a short reflection on a measurement, setup, or programming mistake and how it was corrected.

Never include employer/customer drawings, controlled technical data, export-controlled information, restricted defense data, passwords, internal process sheets, confidential programs, or proprietary part geometry without explicit authorization.

16. Responsible AI, privacy, and cybersecurity

AI can help explain machining concepts, generate study questions, translate unfamiliar terminology, summarize a public manual, or help compare training options. It can also produce unsafe or incorrect technical output.

Safe AI uses

Use AI for:

- practice questions and vocabulary;
- explanations of public concepts you can verify;
- résumé wording based on truthful experience;
- comparison checklists for schools or jobs;
- non-proprietary learning examples;

- language support for public information.

High-risk AI uses

Do not trust AI by itself to provide production-ready:

- G-code;
- toolpaths;
- feeds and speeds;
- workholding decisions;
- offset values;
- dimensional interpretations;
- safety procedures;
- controller recovery instructions;
- inspection acceptance decisions;
- regulated-process documentation.

Treat AI-generated manufacturing instructions as unverified until checked through qualified workplace procedures, simulation, approved programming systems, first-piece controls, and competent supervision.

Protect employer and customer data

Do not upload proprietary drawings, CAD models, CAM files, machine programs, customer names, export-controlled data, medical-device records, aerospace/defense information, passwords, machine credentials, network diagrams, or confidential process information to public AI services without explicit authorization.

Networked CNC controllers and manufacturing systems are computers as well as machines. Follow approved file-transfer, removable-media, account, patching, backup, network-segmentation, and change-control procedures. Do not connect personal devices or software to production equipment without authorization.

17. Accessibility and realistic job fit

Machining can involve standing, reaching, repetitive tasks, close visual inspection, fine measurement, noise, coolant/mist exposure, sharp chips, material handling, and work around moving equipment. Essential functions vary by workplace and role.

Do not self-exclude solely because of age, disability, neurodivergence, or another personal characteristic. Instead, examine the actual job's essential functions, training-lab accessibility, PPE compatibility, communication methods, schedule, lifting expectations, visual/measurement demands, and available accommodation process.

Some people may find adjacent roles more suitable, such as inspection, CAD/CAM, process planning, quoting/estimating, tooling support, documentation, technical training, applications support, or production coordination. These roles still require their own qualifications.

Most hands-on machining work is site-based. Treat “remote CNC operator” claims cautiously. Some programming, process-planning, quoting, applications, or documentation roles may permit remote or hybrid work, but this is employer-dependent and usually easier after substantial manufacturing experience.

18. Career progression

A common progression may look like:

- machine helper / production associate;
- CNC operator;
- setup operator / setup technician;
- machinist;
- advanced or multi-axis machinist;
- CNC programmer / CAM programmer;
- quality inspector / metrology technician;
- toolroom or prototype machinist;
- manufacturing technician or process technician;
- lead, trainer, supervisor, applications specialist, manufacturing engineer technician, or related role depending on education and experience.

Progression is not automatic. Build evidence of safe setup, dimensional control, tooling knowledge, troubleshooting, documentation, cycle improvement, difficult materials, tight tolerances, programming, inspection, and mentoring only as you actually perform those responsibilities.

19. How to compare job offers

Compare more than hourly pay. Ask about:

- shift and schedule;
- overtime expectations and rates;
- shift differentials;
- union status and collective-agreement terms where applicable;
- machines, controllers, materials, part size, and tolerance range;
- operator versus setup versus programming scope;
- training and advancement process;
- production quotas and quality expectations;
- tooling and PPE responsibility;
- coolant, mist, noise, lifting, and ergonomic controls;
- paid breaks, leave, health benefits, retirement/pension, and bonuses;
- whether programming, inspection, setup, or maintenance duties are added without corresponding training or pay;
- commute, transportation, parking, relocation, and worksite stability.

A higher hourly rate may not be a better offer if the schedule, safety culture, unpaid tool costs, unstable hours, or unsupported responsibilities create a worse overall situation.

20. When to pause or decline a pathway

Consider stopping or delaying when:

- a school or recruiter guarantees employment or income;
- you are pressured to borrow before seeing the lab or written costs;
- the provider cannot explain which machines, controllers, lab hours, instructors, and credentials are included;
- a job expects setup/programming/maintenance responsibilities far beyond your demonstrated competence without training or supervision;
- safety guards or interlocks are routinely bypassed;
- scrap, measurements, inspections, or quality records are falsified;
- proprietary or regulated customer data is handled casually;
- required apprenticeship, credential, or funding rules are unclear;
- reimbursement terms are verbal rather than written;
- the physical or schedule demands are incompatible with your needs and reasonable adjustments have not been explored;
- a supposedly “remote” machining job is actually an on-site production role presented misleadingly.

21. Source and verification notes

Primary controlled sources for this revision include:

- U.S. Bureau of Labor Statistics — Machinists and Tool and Die Makers: <https://www.bls.gov/ooh/production/machinists-and-tool-and-die-makers.htm>
- O*NET OnLine — Machinists 51-4041.00: <https://www.onetonline.org/link/details/51-4041.00>
- Apprenticeship.gov — Machinists occupation page: <https://www.apprenticeship.gov/apprenticeship-occupations/listings?occupationCode=51-4041.00>
- Apprenticeship.gov — Advanced Manufacturing: <https://www.apprenticeship.gov/apprenticeship-industries/advanced-manufacturing>
- U.S. Department of Labor — WIOA: <https://www.dol.gov/agencies/eta/wioa>
- OSHA — Machine Guarding: <https://www.osha.gov/machine-guarding/>
- OSHA — Metalworking Fluids: <https://www.osha.gov/metalworking-fluids>
- NIMS — Credentialing: <https://www.nims-skills.org/credentialing>
- NIMS — Renewal Policy: <https://www.nims-skills.org/renewal-policy>
- Canada Red Seal — Machinist: <https://red-seal.ca/eng/trades/machinist.shtml>
- Government of Canada Job Bank — CNC Machinist wages: <https://www.jobbank.gc.ca/marketreport/wages-occupation/16875/ca>

- Canada Apprentice Loan: <https://www.canada.ca/en/services/jobs/training/support-skilled-trades-apprentices/loan.html>
- SENA institutional and employment resources: <https://www.sena.edu.co/> and <https://ape.sena.edu.co/>
- Indeed CNC Machinist salary estimate — **non-government market estimate**: <https://www.indeed.com/career/cnc-machinist/salaries>

Source details, controlled interpretations, and date checks are recorded in `project/revision-2026/guide-33/research/GUIDE_33_CURRENT_SOURCE_EVIDENCE_02.md`.

Links, program availability, wage data, funding rules, and credential requirements change. Verify the current source before making a financial, training, relocation, or employment decision.

22. Final practical checklist

Before pursuing machining/CNC work, confirm that you can answer:

- What exact job level am I targeting: operator, setup, machinist, inspection, or programming?
- Which machines and controllers do local employers use?
- Can I read the drawings and measure the tolerances expected at my target level?
- What safety training and supervision will I receive before operating equipment?
- Is a paid trainee/apprenticeship route available before I borrow for school?
- Have I checked WIOA, public-college, employer, scholarship, Canada apprenticeship, SENA, or other public funding/support options that apply to me?
- Does the training program provide enough supervised hands-on machine time?
- Are any credentials truly required by employers, or merely optional?
- Have I verified current credential validity and renewal rules?
- Am I comparing official wage data separately from non-government salary estimates?
- Do I understand the physical, schedule, noise, coolant, and site-based demands?
- Do I know what data must never be uploaded to public AI systems?
- Can I describe my skills truthfully without claiming setup, programming, inspection, journey person, or credential status I have not earned?

The strongest first step is usually the lowest-cost path that gives you **real supervised machine time, measurement discipline, safety habits, and evidence employers can verify**.